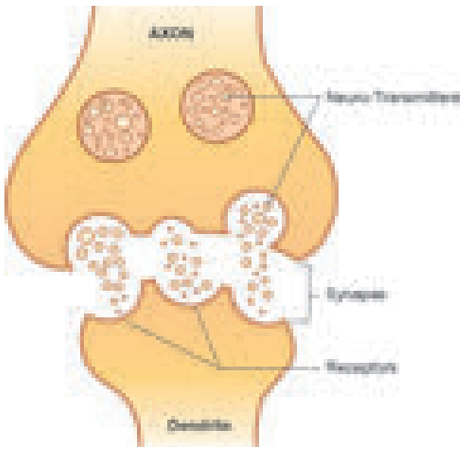


**Fig. 15.3** Nerve impulse transmission

The flow of nerve impulses from axonal end of one neuron to dendrite of another neuron through a **synapse** is called **synaptic transmission**.




Each neuron can transmit 1,000 nerve impulses per second and make as many as ten thousands of synaptic contacts with other neurons.

### 15.2.1 Neurotransmitters

Neurotransmitters are the chemicals which allow the transmission of nerve impulse from the axon terminal of one neuron to the dendron of another neuron or to an **effector organ.** The important neurotransmitter released by neurons is called **Acetylcholine**.

## 15.3 Human Nervous System

The complexity of nervous system can be observed during the course of evolution. We the human beings differ from other animals in our ability to think and take actions, which is due to the well developed nervous system. Human nervous system is differentiated into **central nervous system (CNS), peripheral nervous system (PNS)** and **autonomic nervous system (ANS).**

The CNS acts as centre for information processing and control. It consists of the brain and the spinal cord. The PNS is made up of the nerves which connect the brain and spinal cord to all parts of the body. The ANS is formed of sympathetic and parasympathetic nerves.

### 15.3.1 Central Nervous System

The brain and the spinal cord being delicate vital structures are well protected in bony cavities of the skull and the vertebral column respectively. CNS is formed of two types of matter such as white matter or grey matter with respect to the presence or absence of myelin sheath which we have discussed earlier.

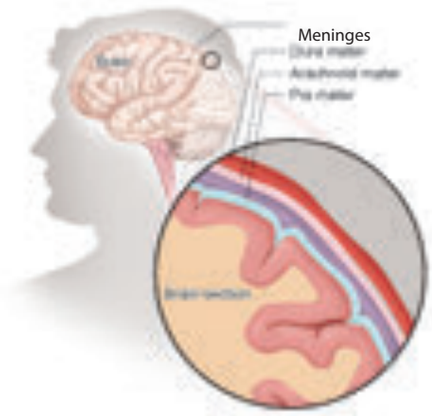



**Fig. 15.4** Meninges of Brain

The brain is the controlling centre of all the body activities. It is covered by three connective tissue membrane or **meninges** :

(i) **Duramater** (*dura:* tough; *mater*: membrane)is the outermost thick fibrous membrane

(ii) **Arachnoid membrane** (*arachnoid*: spider) is the middle, thin vascular membrane providing web like cushion

(iii) **Piamater** (*Pia*: soft or tender) is the innermost, thin delicate membrane richly supplied with blood.

Meningeal membranes protect the brain from mechanical injury.



10th_Science Unit-15.indd 221 11-01-2020 11:26:14